IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

POWER SYSTEM DYNAMICS, STABILITY AND CONTROL

Monday 22 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri

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The Questions

1.

a)

i) What is lightning surge phenomenon in power systems?

[2]

ii) How is this surge characterised in network planning, design and operation?

[1]

iii) How does the lightning surge impact system operation?

[2]

iv) What mitigation measures are in place to address the impact of a lightning surge in system operation?

[2]

b)

i) What is sub-synchronous resonance (SSR) in the context of power system operation?

[4]

ii) How sub synchronous torsional interaction (SSTI) is different from sub-synchronous control interaction (SSCI)?

[2]

iii) How has the nature of SSR evolved with more converter interfaced generation (CIG) in electric power systems?

[2]

iv) List two approaches to controlling SSR

[2]

c)

i) What is *transient stability* in the context of classical power system operation?

[3]

ii) How does one improve the transient stability of a power system?

[2]

iii) How has the concern of transient stability changed in view of more renewable generation in the system?

[3]

[25 marks]

2.

a)

i) Why are the rotors of synchronous generators round shaped for the power generation application in thermal and nuclear power plants?

[3]

ii) Usually the stator of a synchronous generator is laminated whereas the rotor is solid: explain the reason.

[4]

iii) Synchronous machines have synchronous reactance in the range of 1.00 to 2.0 p.u. During a solid fault at the terminal of a synchronous generator, the initial fault current can be as high as 6-8 times the normal load current despite the large synchronous reactance. Explain the reason with reasonable details.

[4]

b) A round rotor synchronous generator is connected to the grid at 23 kV (line to line) voltage. The rated power of 500 MW, power factor is 0.85 (lagging) . The synchronous reactance is 1.25 p.u. on its rated base of 588 MVA. The resistance is negligible. Find out:

i) the rated current in kA

[2]

ii) the excitation (line to line) voltage in kV

[3]

iii) load angle (δ) in degree

[2]

iv) stability margin in degree

[1]

iv) reactive power in MVAR

[3]

c) The excitation voltage is increased by 10%

i) Find out the change in stability margin in degree with respect to the rated excitation voltage.

[3]

[25 marks]

3.

a)

i) What is Park transformation in the context of synchronous machine dynamic modelling ?

[2]

ii) How does Park transformation overcome the limitation of 'abc' stationary reference frame for circuit analysis?

[4]

iii) What are stator transients in the contexts of circuit model of synchronous machine dynamics ?

[3]

iv) Why are such transients neglected in transient stability studies?

[3]

v) What will be the impact of the inclusion of stator transients on the generator's dynamic behaviour ?

[3]

b)

i) What are the basic functional and constructional differences between the Type 2 (WRIG) and Type 3 (DFIG) wind turbine generators ?

[4]

ii) What is Tip Speed Ratio (TSR) in the context of wind turbine operation ?

[3]

iii) How does TSR influence the power coefficient (Cp) of a wind turbine ?

[3]

[25 marks]

4.

a)

i) What is the Betz limit in wind power conversion?

[2]

ii) Starting from the first principle of aerodynamics, establish that the Betz limit is 16/27.

[7]

b)

i) What is passive stall in a wind turbine?

[2]

ii) How is such stall achieved in practice?

[2]

iii) How does active stall control differ from passive stall control?

[2]

iv) Why has the industry discontinued active stall control?

[2]

v) How is the gearbox modelled for dynamic studies of wind turbines ?

[4]

vi) Why are turbine blades not controlled below the rated wind speed ?

[2]

vii) Why does a large off-shore wind farm, located deep in the sea, requires HVDC technology for power transmission ?

[2]

[25 marks]